

# NLP HW3 — Q2: WhoDunnit (Multi-Agent Systems)

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## 2.1 Naive Agent Implementation (Analysis)

We ran the naive baseline with the default configuration and no additional flags (`python run_game.py --iterations 3`: 3 suspects, `max_turn_count = 11`, no summarizer, no internal thought). *Alex* is the Accuser and *Beth* the Intel agent. The full log is in Appendix ??; the analysis below cites it directly.

**What specific behavioral problems did you identify in the initial run of the naive agents? Provide at least two concrete examples from your `game_log`.**

**Problem A — Analysis paralysis:** the Accuser will not accuse even when the culprit is already uniquely determined.

- **Game 1** (culprit = Suspect 3): only Suspect 3 wears square glasses, so Beth's answer `square eye glasses` → `Suspect 3` already fixed the culprit. Alex still asked four more broad questions and never accused, hitting the turn limit.
- **Game 2** (culprit = Suspect 3): `blue eye color` → S2, S3 and `circular eye glasses` → S1, S3 intersect to {S3}, yet Alex kept questioning until timeout.

**Problem B — No cross-turn reasoning.** The Accuser treats each answer in isolation and never intersects the narrowing suspect lists into a running candidate set, so it cannot recognize that only one suspect remains — the root cause of Problem A.

**Do the agents dynamically utilize the full spectrum of their available action space, or do they prematurely converge on a single, repetitive action type? Analyze how this behavioral pattern affects their overall progress.**

They **converge**. Alex issues only `request-broad` in all three games (zero `request-specific`; `accuse` only once, in Game 3), which in turn forces Beth into `respond-broad` only. Because the terminal `accuse` action is under-used, information accumulates but is rarely acted upon, so games stall: Games 1 and 2 exhaust all 11 turns without an accusation, ending in timeout rather than a decision.

**Does the Accuser make a blind accusation prematurely, or does it get stuck in an endless loop until the `max-turn-count` is reached? Why do you think it struggles to finalize the decision?**

It gets stuck in an **endless questioning loop**, not a premature accusation: Games 1 and 2 reach `max-turn-count` with no accusation, and Game 3 accuses correctly only after redundant confirmation. It struggles to finalize because:

1. **Stateless prompting** — no maintained candidate set, so it never sees that one suspect remains;
2. the naive prompt **forbids reasoning**, so the model defaults to asking one more question rather than committing;

3. no firm stopping criterion, so “accuse when certain” never triggers.

## A Naive baseline game log

```
Game 1
Suspects:
Suspect 1: their eye color is brown, they have circular eye glasses, their hair is long, is wearing a black
    hat, their hobby is basketball, is in a happy mood, they have white shoes, they have a bronze watch, they
    have long, black pants and a pink, button-up shirt.
Suspect 2: their eye color is green, they have circular eye glasses, their hair is long, is wearing a brown
    hat, their hobby is basketball, is in a happy mood, they have red shoes, they have a silver watch, they
    have short, black pants and a pink, button-up shirt.
Suspect 3: their eye color is green, they have square eye glasses, their hair is long, is wearing a black hat,
    their hobby is dancing, is in a sad mood, they have red shoes, they have a silver watch, they have long,
    black pants and a green, tee shirt.
==
The culprit is: 3
==
Communication channel:
Alex asked a broad question about: green eye color
Beth responded with: Suspect 2, Suspect 3
Alex asked a broad question about: square eye glasses
Beth responded with: Suspect 3
Alex asked a broad question about: long hair
Beth responded with: Suspect 1, Suspect 2, Suspect 3
Alex asked a broad question about: black hat
Beth responded with: Suspect 1, Suspect 3
Alex asked a broad question about: red shoes
Beth responded with: Suspect 2, Suspect 3
Alex asked a broad question about: silver watch

=====
Game 2
Suspects:
Suspect 1: their eye color is green, they have circular eye glasses, their hair is short, is wearing a black
    hat, their hobby is basketball, is in a happy mood, they have white shoes, they have a bronze watch, they
    have short, black pants and a pink, tee shirt.
Suspect 2: their eye color is blue, they have square eye glasses, their hair is short, is wearing a brown hat,
    their hobby is dancing, is in a happy mood, they have red shoes, they have a silver watch, they have
    short, black pants and a green, tee shirt.
Suspect 3: their eye color is blue, they have circular eye glasses, their hair is short, is wearing a brown
    hat, their hobby is dancing, is in a happy mood, they have white shoes, they have a silver watch, they
    have short, brown pants and a green, tee shirt.
==
The culprit is: 3
==
Communication channel:
Alex asked a broad question about: blue eye color
Beth responded with: Suspect 2, Suspect 3
Alex asked a broad question about: circular eye glasses
Beth responded with: Suspect 1, Suspect 3
Alex asked a broad question about: short hair
Beth responded with: Suspect 1, Suspect 2, Suspect 3
Alex asked a broad question about: brown hat
Beth responded with: Suspect 2, Suspect 3
Alex asked a broad question about: white shoes
Beth responded with: Suspect 1, Suspect 3
Alex asked a broad question about: silver watch

=====
Game 3
Suspects:
Suspect 1: their eye color is brown, they have square eye glasses, their hair is short, is wearing a black hat
    , their hobby is basketball, is in a sad mood, they have red shoes, they have a bronze watch, they have
    short, black pants and a green, button-up shirt.
```

Suspect 2: their eye color is green, they have circular eye glasses, their hair is long, is wearing a brown hat, their hobby is dancing, is in a sad mood, they have red shoes, they have a bronze watch, they have short, black pants and a pink, button-up shirt.

Suspect 3: their eye color is brown, they have circular eye glasses, their hair is long, is wearing a brown hat, their hobby is basketball, is in a happy mood, they have red shoes, they have a bronze watch, they have short, black pants and a pink, tee shirt.

==

The culprit is: 1

==

Communication channel:

Alex asked a broad question about: brown eye color

Beth responded with: Suspect 1, Suspect 3

Alex asked a broad question about: square eye glasses

Beth responded with: Suspect 1

Alex asked a broad question about: short hair

Beth responded with: Suspect 1

Alex asked a broad question about: black hat

Beth responded with: Suspect 1

Alex accused Suspect 1 and was correct!

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